

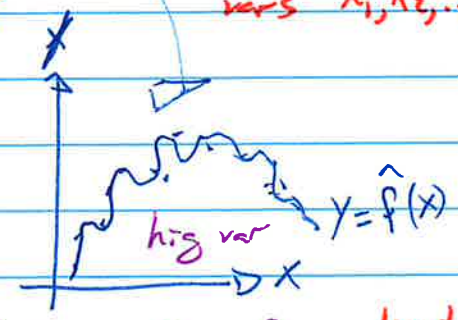
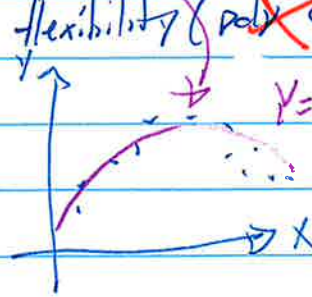
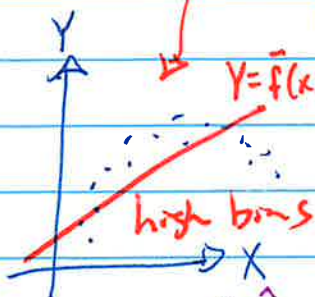
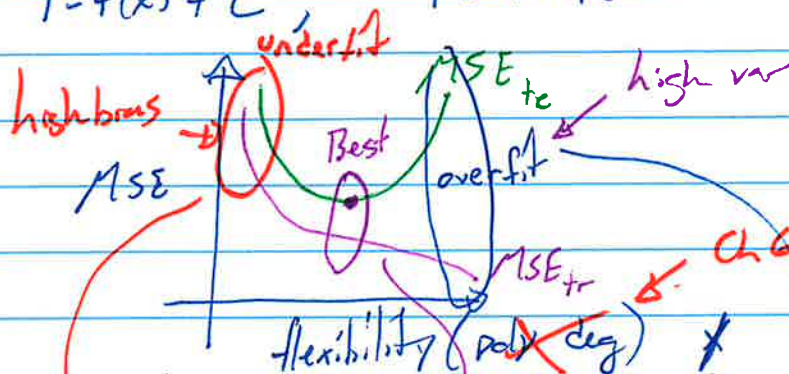
Ch 6 Linear model selection & regularization

Idea: Handle special case of overfitting.
↳ Solution: Bias-variance tradeoff. Top concept!

1pg fast sketch of key ideas of Bias-variance tradeoff
Try it!

Bias-variance tradeoff tackles over/under fitting.

$$Y = f(x) + \varepsilon, \quad Y \approx \hat{Y} = \hat{f}(x)$$



$$E[(y_0 - \hat{f}(x_0))^2] = \text{Bias}(\hat{f}(x_0))^2 + \text{Var}(\hat{f}(x_0)) + \text{Var}(\varepsilon)$$

$$\text{Bias}(\hat{f}(x_0)) = E[(\hat{f}(x_0))^2] - f(x_0)$$

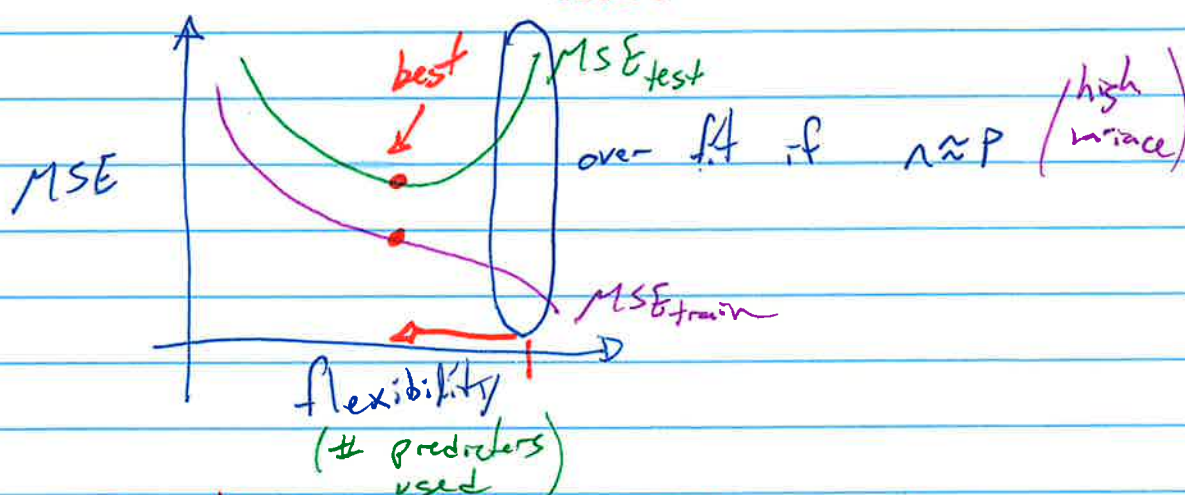
$$V_{\mu}(\hat{f}(x_0)) = E[(\hat{f}(x_0) - \mu)^2], \mu = E[\hat{f}(x_0)]$$

Ch 6 focus on ^{assume} multiple linear regression

$$Y = f(x) + \varepsilon \implies (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$$

in the case of overfitting. ($n \approx p$) ^{Ch 3} _{small train dataset} ^{many predictors}

Bias-variance tradeoff:



Decrease flexibility & find best model
 2 ways: ① Best subset of variables (some of $X_1, X_2, \dots, X_p$)
 ② Coef β_i shrinkage

Best subset of variables

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon \quad (n \approx p, \text{overfit})$$

Remove some X_i 's. to find best model.

- 3 ways: ① Check all possible models (best subset sel)
 $\hookrightarrow$ net.
 ② Forward sel
 ③ Backwards sel } - more eff.

Best subset selection Train & assess all possible models.

Ex $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$. M_0 $\boxed{Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon}$ OLS on all 3 vars (OLS)

Possible var choices:

	1 var	2 var	3 var OLS
None (M_0)	$X_1 (M_1)$	$X_1 + X_2 (M_4)$	
P	$X_2 (M_2)$	$X_1 + X_3 (M_5)$	$X_1 + X_2 + X_3 (M_7)$
null model	$X_3 (M_3)$	$X_2 + X_3 (M_6)$	

Total of 8 models: $8 = 2^3$ (count ways to include X_i / yes/no)

For $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$, need

2^p total models.

p	10	20	30
2^p	$2^{10} = 1024$	$2^{20} \approx 1,000,000$	1 bil

If $p > 10$, not an option.

Forward selection Add one variable at a time if has best increase.

Ex $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$. M_0 (null model)

$X_1 \rightarrow M_1$: Best of all single var models

$X_1 + X_2 \rightarrow M_2$: Use X_1 & best of all 2 var

M_3 : Repeat models

$\begin{matrix} X_1 \\ X_2 \\ X_3 \\ X_1 \\ X_2 \\ X_3 \end{matrix}$

Models train: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$

$p+1$ "best" models, pick top one.

$M_0, M_1, M_2, \dots, M_p$
 (1) (p ways) (p-1 ways) (1 way)

$\sim p^2$
 $\uparrow$

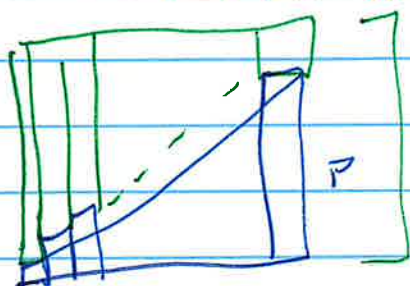
Total models trained:

$1 + \frac{p(p+1)}{2}$
 models

$$1 + p + (p-1) + (p-2) + \dots + 2 + 1 = \frac{p(p+1)}{2}$$

(triangular #s)

$$1 + 2 + 3 + 4 + \dots + (p-1) + p$$



$$p(p+1)$$

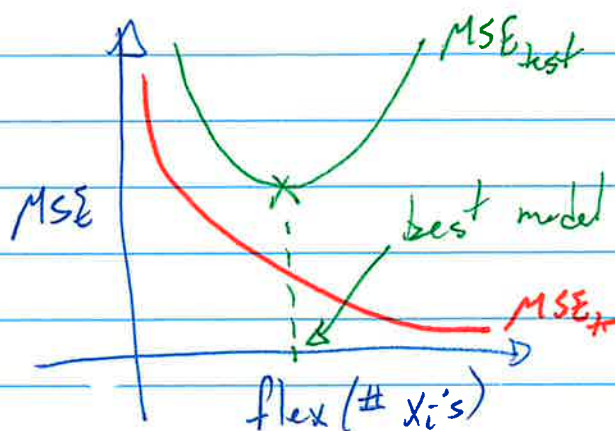
Around p^2 models trained, much less than p^p

Note: F.S. models are subset of B.S.S. & best models may not match.

Backwards Selection: Similar to F.S., but start with all variables & remove one at a time which yields worst model. Counts match F.S. but may have different "best" models.

How to choose best model?

Cannot use MSE_{tr} to pick best model.



Instead, approx MSE_{test} & choose min. 2 ways

① CV (k-fold)

↳ on all best models

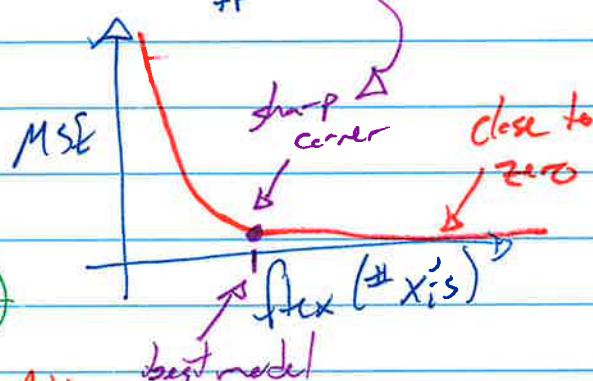
• BS (2^p total CV)

• FS/BS ($\approx p^2$ to CV)

② Stats which approx

MSE_{test} via MSE_{train}

Note, sometimes use elbow method on MSE_{tr} or R^2_{tr}



Info criterion

$$\begin{bmatrix} C_p \\ AIC \\ BIC \end{bmatrix} \quad \frac{1}{n} (RSS + \text{correction})$$

$$RSS / (n - \text{\# vars used})$$

Adds df

As d incs, R^2 dec

Adj $R^2 = 1 - \frac{RSS / (n - 1)}{TSS / (n - 1)}$

Recall: $R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} \in [0, 1]$

% var expl by model

If stats or CV disagree, best practice is

- ① Simple models best (few X_i 's as possible)

- ② More CV on top candidates

- ③ One-standard error rule / simplest one SE from best per stat

Ch 6 Variable selection & regularization

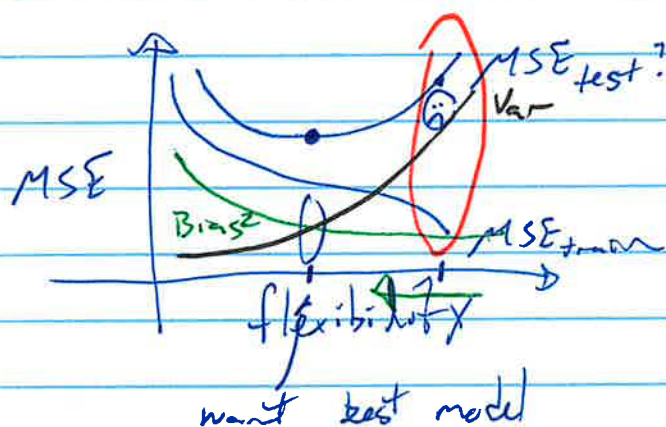
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pg 1

M.L.R: $Y = f(x) + \varepsilon \stackrel{\text{assume}}{=} (\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p) + \varepsilon$

Issue: Overfitting ($p \approx n$)

↳ High variance of $\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$ OLS



2 strategies:

- ① Variable selection (only some of $x_1, \dots, x_p$)
 - ↳ 3 ways: Best subset, forward sel, back sel
 - ↳ pick best model
 - Complexities: $\mathcal{O}(2^p)$, $\mathcal{O}(n^2)$, $\mathcal{O}(n^2)$
 - ① MSE_{test} stats ($C_p, AIC, BIC, adj R^2$)
 - ② CV (*)

② β_j coef shrinkage.

↳ 2 ways: ① Ridge regression

② LASSO (least abs shrinkage + selection operator)

Ridge Regression: $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon$

(Ch 3) OLS: $\vec{\beta} = \min_{\beta_j} RSS = \min_{\beta_j} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

$$= \min_{\beta_j} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

(*)

Force $\|\vec{\beta}\|$ small λ^2

Ridge reg

$$\vec{\beta}_j^R = \min_{\vec{\beta}_j} \left(\text{RSS} + \text{shrinkage} \right)$$

$$= \min_{\vec{\beta}_j} \left(\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2 \right)$$

not for β_0

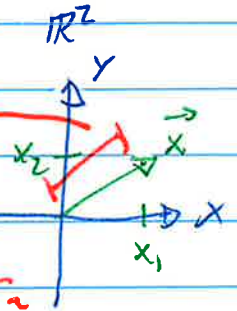
where λ^2 is parameter via CV.

shrinkage term: $\lambda \sum_{j=1}^p \beta_j^2$

Math 308

$$\|\vec{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

(Euclidean norm)
 ℓ^2 -norm



λ -role:

$\lambda = 0$

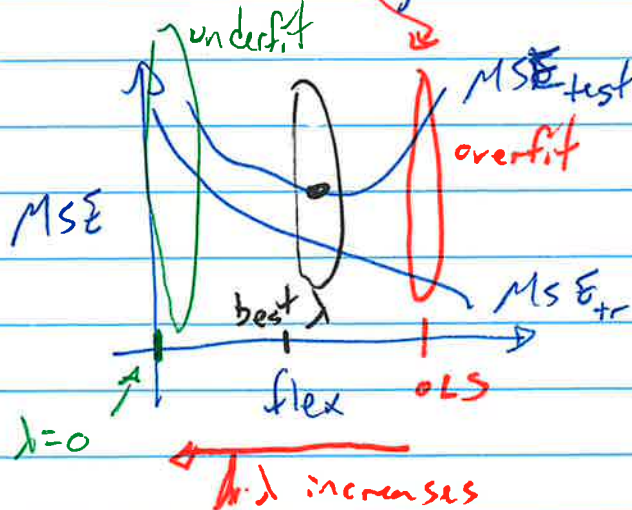
OLS (no shrink)

$\|\vec{x}\| = \sqrt{x_1^2 + x_2^2}$

$0 < \lambda < \infty$ in between

$\lambda \rightarrow \infty \quad \beta_j \rightarrow 0 \quad j \geq 1$

gives null ($y = \beta_0 + \epsilon$) model



Note: Need computer alg to fit λ 's (gradient descent)

Warning OLS gives same $\hat{\beta}$ without normalizing X_i 's. $\hat{\beta}_i$ interpretable pg 3

Ridge regression is sensitive to X_i scales, need to standardize

(z-score) for X_i : $Z_i = \frac{X_i - \bar{X}}{S}$ $\bar{X} = \frac{1}{n}(X_1 + \dots + X_n)$ $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$

$\bar{X}$ mean S stand dev

mean 0 & SD 1 All Z_i 's on same scale.

Result is standardized $\hat{\beta}_{RR}$, can unstandardize.

Comments: ① Ridge regression has adv over variable selection of efficiency. (only 1-cv), though maybe less interpretable.

② $n < p$, OLS cannot be fit but RR still can be.

LASSO Alt approach to RR

$$\hat{\vec{\beta}}_{\lambda}^L = \min_{\beta_j} \left(\underbrace{\sum_{i=1}^n (y_i - \beta_0 + \sum_{j=1}^p \beta_j x_{ij})^2}_{RSS} + \lambda \underbrace{\sum_{j=1}^p |\beta_j|}_{\text{Shrinkage}} \right)$$

Main diff use $\|\vec{x}\|_1$ instead of $\|\vec{x}\|_2$
 $\|\vec{x}\|_1 = |x_1| + |x_2| + \dots + |x_n| = \sum_{i=1}^n |x_i|$

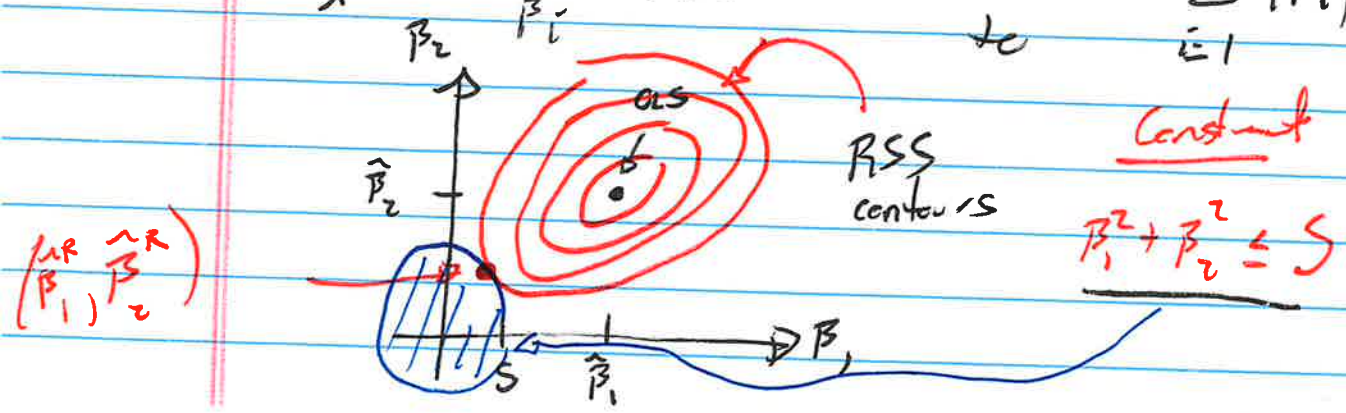
Choose best $\lambda \geq 0$ via CV.

Strangeness: LASSO makes some coefs 0 as $\lambda \rightarrow \infty$ (does var selection)

Main reason: Reformulate minimization into constrained optimization.

RR $\hat{\vec{\beta}}_{\lambda}^R = \min_{\beta_i} \overset{OLS}{RSS}$ subject to $\sum_{i=1}^p \beta_i^2 \leq \underset{R}{S(\lambda)}$

LASS $\hat{\vec{\beta}}_{\lambda}^L = \min_{\beta_i} RSS$ subject to $\sum_{i=1}^p |\beta_i| \leq \underset{L}{S_L(\lambda)}$

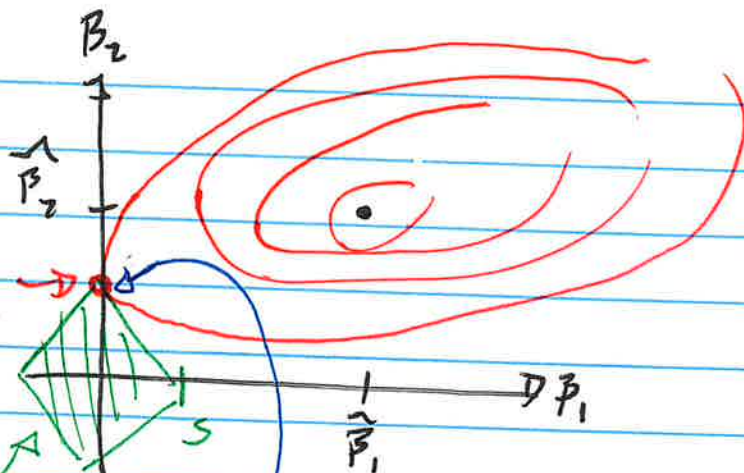


LASSO:

Corners
are
most
likely
hit.

$$|\hat{\beta}_1| + |\hat{\beta}_2| \leq S$$

$$\hat{\beta}_1^L = 0$$



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